Warm-up / A chance of arriving late

Make sense of a chance forecast

Name: _____ Date: _____

Suggested lesson time: 30 minutes. Work individually, then compare when invited.

Useful words

Probability: A stated chance that a defined event will occur.

Calibration: A match between stated chances and outcome frequencies across many comparable forecasts.

Comparable forecasts: Forecasts made for similar events, conditions, and time horizons.

Read the scenario

Invented scenario: A travel app gives tomorrow's bus a 25% chance of reaching the terminal after 9 a.m. Imagine 100 comparable forecasts made one day ahead at 25% by a well-calibrated app: its chances match long-run frequencies.

Produce: Write two approximate counts and one sentence about the chance of tomorrow being late.

Your steps

1. State about how many of the 100 buses should arrive after 9 a.m. and how many should not.

2. Explain why a 25% forecast does not rule out a late bus tomorrow.

Before you finish: Keep the event “after 9 a.m.” clear; do not turn “unlikely” into “impossible”.

Your response

Main activity / What does 70% rain mean?

Name: _____

Read the scenario

Invented scenario: A weather app assigns a 70% chance of measurable rain tomorrow. Here, measurable rain means at least 1 mm at the town weather station during that calendar day. Imagine 100 comparable forecasts, each issued one day ahead at 70%, from a well-calibrated app: its stated chances match long-run frequencies.

Produce: Write two approximate counts, an explanation of one dry day, and a plan for checking the forecast record.

Your steps

1. Write approximately how many of those 100 days should have measurable rain and how many should not.
2. Explain why a dry tomorrow would not by itself refute a 70% forecast.
3. Describe the record you would inspect to judge whether the app's 70% forecasts are well calibrated.

Before you finish: Do not treat 70% as a guarantee, 70% of the town, or 70% of the day.

Your response

Exit ticket / A delivery forecast record

Name: _____

Read the scenario

Invented scenario: A courier predicts an 80% chance that a parcel will arrive by 6 p.m. tomorrow. Its record contains 200 comparable one-day-ahead forecasts made at 80%; 160 parcels arrived by 6 p.m. and 40 did not.

Produce: Write a frequency calculation and two sentences judging the record and a late parcel.

Your steps

1. Calculate the on-time percentage: on-time parcels / all parcels x 100.
2. State whether this record matches the stated 80% chance, without claiming it guarantees future accuracy.
3. Explain why one late parcel would not by itself show an 80% forecast was wrong.

Before you finish: Include all 200 forecasts and preserve the possibility of a late parcel.

Your response